Week 1

Machine Learning Fundamentals

Classification · Statistics · Dimensionality · Distributions · Monitoring

Contents

01

Classification Metrics & Accuracy

After training a classification model, we need to evaluate how good it is. For classification problems, the model predicts categories (Spam/Not Spam, Fraud/Not Fraud, Disease/No Disease). The simplest metric is **Accuracy**.

```
Accuracy = (TP + TN) / (TP + TN + FP + FN)
```

Where: TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative

Example: 90 correct predictions out of 100 observations → Accuracy = 90%

Accuracy works for multi-class too. 85 correct out of 100 (Cat/Dog/Horse) → 85%.

How much accuracy is good?

Context matters

There is no universal rule. 70% may be excellent for some problems; 99% may still be poor for others.

Movie recommendation

80% accuracy may be perfectly acceptable.

Cancer detection

80% accuracy is dangerously insufficient.

Scikit-Learn usage

Use accuracy_score(y_true, y_pred) to compute accuracy in Python.

Warning — Imbalanced dataset trap: A dataset with 990 negatives and 10 positives will give 99% accuracy if the model always predicts 'Negative' — yet it never detects a single positive case. This model is completely useless despite its high accuracy score.

02

Confusion Matrix & Error Types

The confusion matrix shows what the model predicted vs. what the actual values were. It solves the limitations of accuracy by revealing exactly which classes are confused.

Actual \ Predicted	Positive	Negative
Positive	**TP — True Positive (Correct)**	**FN — False Negative (Type II Error)**
Negative	**FP — False Positive (Type I Error)**	**TN — True Negative (Correct)**

True Positive (TP)

Actual = Positive, Predicted = Positive. Patient has disease, model predicts disease. Correct.

True Negative (TN)

Actual = Negative, Predicted = Negative. Healthy patient, model predicts healthy. Correct.

False Positive (FP) — Type I

Actual = Negative, Predicted = Positive. Healthy patient diagnosed sick. Legitimate email flagged as spam.

False Negative (FN) — Type II

Actual = Positive, Predicted = Negative. Sick patient diagnosed healthy. Fraud passes undetected.

Which error is worse? It depends on the application. In disease detection, Type II (missing disease) is usually worse — life-threatening. In spam filtering, Type I (losing important emails) often hurts more. In fraud detection, Type II (undetected fraud) is more costly.

Multi-class confusion matrix

For N classes, the matrix becomes N×N. Diagonal = correct predictions. Off-diagonal = mistakes.

Actual \ Pred	Cat	Dog	Horse
Cat	✓ Correct	✗ Wrong	✗ Wrong
Dog	✗ Wrong	✓ Correct	✗ Wrong
Horse	✗ Wrong	✗ Wrong	✓ Correct

03

Dimensionality Reduction

A **dimension** simply means a feature or variable. 10 features = 10-dimensional space. Higher dimensions create unexpected mathematical problems.

Curse of Dimensionality

As dimensions increase, data space grows exponentially, points become sparse, and distances become meaningless. To maintain the same density:

```
1D: 10 points | 2D: 100 points | 3D: 1,000 points | 10D: 10,000,000,000 points
```

Data sparsity

100 points in 1D are close together. The same 100 points in 100D become extremely far apart.

Distance meaningless

In high dimensions, the difference between nearest and farthest neighbor shrinks toward zero.

KNN degradation

KNN calculates distances. In high dimensions, all points seem equally far — predictions collapse.

Overfitting

High dimensions = model memorizes noise. High training accuracy, low test accuracy.

Hughes phenomenon

Adding features initially improves accuracy, then after a peak it hurts performance.

Feature quality > quantity

Extra features often introduce noise, redundancy, and irrelevance that confuse the model.

Solutions

Feature selection

Keep only important features. Remove irrelevant, duplicate, weak predictors. Methods: Correlation Analysis, Mutual Information, Chi-Square Test.

PCA

Principal Component Analysis — creates new dimensions preserving maximum variance.

LDA

Linear Discriminant Analysis — projects data while maximizing class separation.

t-SNE

t-distributed Stochastic Neighbor Embedding — best for visualizing high-dimensional data in 2D/3D.

Real-life example: A 100×100 grayscale image has 10,000 features (dimensions). A colour image has 30,000. Without dimensionality reduction, training becomes very expensive.

04

Statistics Basics & Variable Types

Descriptive statistics

Summarizes and describes data (mean, median, mode, variance, SD). Uses tables, histograms, pie charts, box plots. Does NOT make predictions.

Inferential statistics

Draws conclusions about a population using sample data. Enables hypothesis testing. E.g., estimate all 20,000 students' GPA from a sample of 100.

Types of Variables

Numerical — Discrete

Countable values. No decimals. Examples: number of children (0,1,2...), cars in a parking lot.

Numerical — Continuous

Any value in a range including decimals. Examples: height (170.5 cm), temperature, salary.

Categorical — Nominal

Categories with no natural order. Examples: blood group, gender, city, country.

Categorical — Ordinal

Categories with meaningful order. Examples: education level, satisfaction (Poor < Good < Excellent).

ML preprocessing: Numerical → Scaling/Normalization/Standardization. Categorical → Label Encoding or One-Hot Encoding. Models cannot directly understand 'Red', 'Blue', 'Green' — must convert to numbers.

05

Variance & Standard Deviation

Two datasets can have the same mean but very different distributions. **Dispersion** measures how spread out data is around the mean.

```
Dataset A: 10, 10, 10, 10, 10 → Mean = 10, SD ≈ 0 Dataset B: 1, 5, 10, 15, 19
→ Mean = 10, SD ≈ 6.6
```

Formulas

```
Population Variance: sigma^2 = Sum(xi - mu)^2 / N Sample Variance: s^2 = Sum(xi -
x_bar)^2 / (n-1) [Bessel's correction] Standard Deviation: sigma = sqrt(sigma^2)
```

Worked example: {2, 4, 6, 8, 10}

Step	Calculation	Result
1 — Mean	(2+4+6+8+10) / 5	x■ = 6
2 — Deviations	–4, –2, 0, +2, +4	Sum = 0 (cancel out)
3 — Squared	16, 4, 0, 4, 16	Sum = 40
4 — Variance	40 / 5	$\sigma^2 = 8$
5 — Std Dev	$\sqrt{8}$	$\sigma \approx 2.83$

Uses in ML: Data analysis, outlier detection (values far from mean are often outliers), Z-score standardization before training many algorithms, probability distributions.

INTERVIEW Q&A;

Why do we square deviations while calculating variance?

Because if we simply add deviations from the mean, positive and negative values cancel each other and the result becomes zero. Squaring makes all values positive and gives a meaningful measure of spread.

06

Kolmogorov-Smirnov (KS) Test

A **non-parametric** statistical test that checks whether a sample follows a specified distribution (one-sample) or whether two samples come from the same distribution (two-sample). Does not require assumptions about population parameters.

```
KS Statistic: D = max | Fo(x) - Fe(x) | Critical value: D_alpha = 1.36 /
sqrt(n) [for alpha = 0.05] Decision: Dcal > Dtab → Reject H0 (distributions
differ) Dcal <= Dtab → Accept H0 (distributions same)
```

One-sample KS test

Checks if a sample follows a given distribution (e.g., normal). H0: sample follows the distribution.

Two-sample KS test

Compares two independent samples. H0: both come from the same population.

Advantages

No normality assumption. Simple to apply. Works on small samples.

Limitations

Less powerful than parametric tests when normality holds. More sensitive at center than tails.

INTERVIEW Q&A;

What is the Kolmogorov-Smirnov Test?

The KS Test is a non-parametric test used to determine whether a sample follows a specified distribution or whether two samples come from the same distribution. It is based on the maximum absolute difference between cumulative distribution functions. If Dcal > Dtab, the null hypothesis is rejected.

07

Model Monitoring & Drift

A deployed model is not 'train once and forget'. Real-world data distributions shift over time, causing model performance to degrade. This is called **drift**.

Data drift (covariate)

Input feature distribution changes. $P_{train}(X) \neq P_{prod}(X)$. E.g., avg salary shifts from ■30k to ■70k.

Concept drift

Relationship between input and output changes. $P(Y|X)$ shifts. E.g., high income used to mean low default risk — post-recession it doesn't.

Prediction drift

Distribution of model predictions changes. Fraud scores that were 0.1–0.3 are now 0.7–0.9.

What to monitor

Input features

Monitor mean, median, variance, missing values, and feature distributions.

Predictions

Monitor prediction distributions, confidence scores, probability scores.

Model performance

Track accuracy, precision, recall, F1 score, ROC-AUC over time.

System metrics

Latency, throughput, CPU usage, memory usage (MLOps infrastructure).

Drift detection methods

Method	Description
KS Test	$\max|F_1(x)–F_2(x)|$. Higher value = larger distribution difference.
PSI (Population Stability Index)	Popular in banking. PSI <0.1 = stable, 0.1–0.25 = moderate, >0.25 = significant.
Jensen-Shannon Distance	Measures similarity between probability distributions.
Wasserstein Distance	Measures how much one distribution must shift to match another. Best for continuous va

Tools: Evidently AI, MLflow, Prometheus, Grafana, WhyLabs, Fiddler AI

08

Normal Distribution & Empirical Rule

A **Normal (Gaussian) Distribution** is bell-shaped, symmetric around its mean. Most observations cluster near the center. Defined entirely by **mean (μ)** and **standard deviation (σ)**.

Bell-shaped curve

Symmetric shape — left and right sides are mirror images.

Mean = Median = Mode

All three central tendency measures are equal and at the center.

Total area = 1

The entire probability under the curve equals 100%.

PDF formula

$f(x) = (1 / \sigma\sqrt{2\pi}) \times e^{(-(x-\mu)^2 / 2\sigma^2)}$

Empirical Rule (68–95–99.7)

Range	Data covered	Example (mean=70, SD=10)
$\mu \pm 1\sigma$	≈ 68%	Scores between 60 and 80
$\mu \pm 2\sigma$	≈ 95%	Scores between 50 and 90
$\mu \pm 3\sigma$	≈ 99.7%	Scores between 40 and 100

Z-Score (Standard Normal)

```
z = (x − μ) / σ
```

Converts any normal variable to standard normal ($\mu=0$, $\sigma=1$). Tells how many standard deviations a value is from the mean. Used for standardization, outlier detection ($|z| > 3$ = potential outlier), and anomaly detection.

Values outside $\mu \pm 3\sigma$ ($|z| > 3$) are considered unusual or potential outliers. The empirical rule provides a quick mental model for understanding data spread.

09

Central Limit Theorem (CLT)

If you repeatedly take samples from **any** distribution and compute their averages, those averages will form an approximately **normal distribution** — regardless of what the original data looks like.

```
Process: Any distribution → Take samples → Compute mean → Repeat many times
Result: Distribution of sample means → Normal (bell curve)
```

Mathematical version

```
X1, X2, ..., Xn are i.i.d. random variables with mean μ and variance σ² Then:
(X_bar − μ) / (σ / sqrt(n)) → N(0, 1) as n → ∞ Variance of sample mean:
Var(X_bar) = σ² / n → Bigger sample = less fluctuation = tighter bell curve
```

Why it works

A sample mean is a sum of random variables. Addition smooths randomness — extreme values cancel out, small fluctuations average out.

What CLT says

The distribution of sample means becomes approximately normal as n grows, for any original distribution with finite mean and variance.

What CLT does NOT say

The original data becomes normal. Individual observations become normal. Only sample means converge to normality.

Assumptions

Samples must be independent. Same underlying process (i.i.d.). Finite mean and variance.

Real-life example

A single dice roll is uniform (1–6). Roll 10 dice and average them, repeat thousands of times → averages form a normal distribution.

Why it matters in ML

Foundation for confidence intervals, hypothesis testing, A/B testing, error estimation, and sampling theory.

INTERVIEW Q&A;

What is the Central Limit Theorem?

The CLT states that the distribution of sample means of independent random variables approaches a normal distribution as sample size increases, regardless of the original distribution, provided the variables have finite mean and variance.